What distinguishes the ways undergrads, graduate students, and new-grads spend their time?

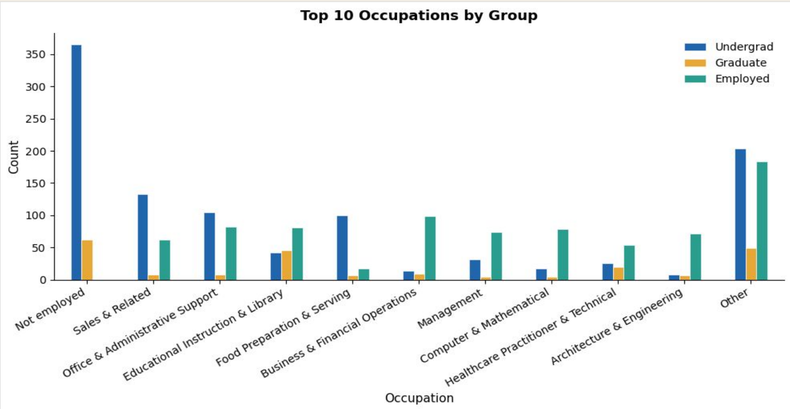
Alisha Li – Spring 2026 Data Science Capstone

WELLESLEY Data Science

Background & Motivation

Lifestyle is a key factor in undergrads' decisions on what to pursue after college. While prior research primarily focuses on time use within each group, little work has been done comparing between the groups. This study aims to see if there are broad differences across the three groups, which may prompt future investigations.

About the data

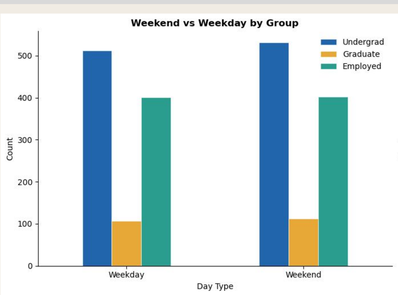


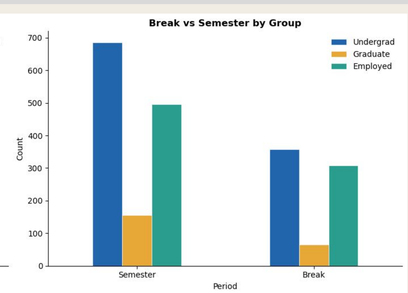
American Time Use Survey

- Participants recall how their spent their prior day starting from 4am to 4am
- 2015-19, 2023-24
- Ages 19-26

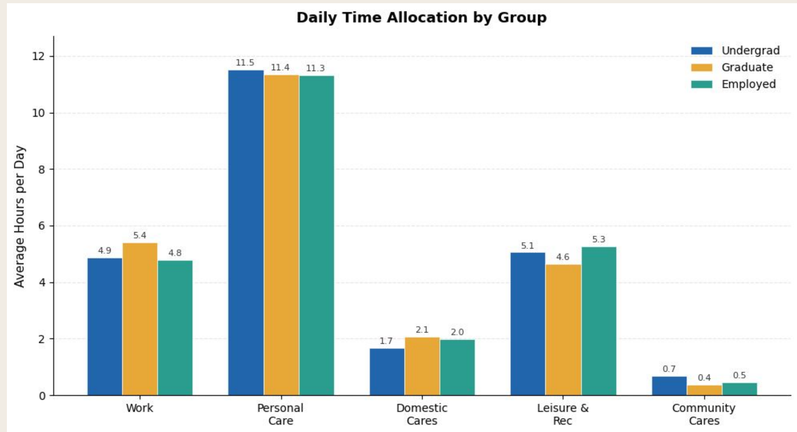
Final Dataset

- 2067 respondents (1044 Undergrad, 803 Employed, 220 Graduate)
- 3 background variables: sex, year, weekend, break
- 6 activity buckets: work, personal care, domestic cares, leisure/rec, community cares, NA





Exploratory Data Analysis

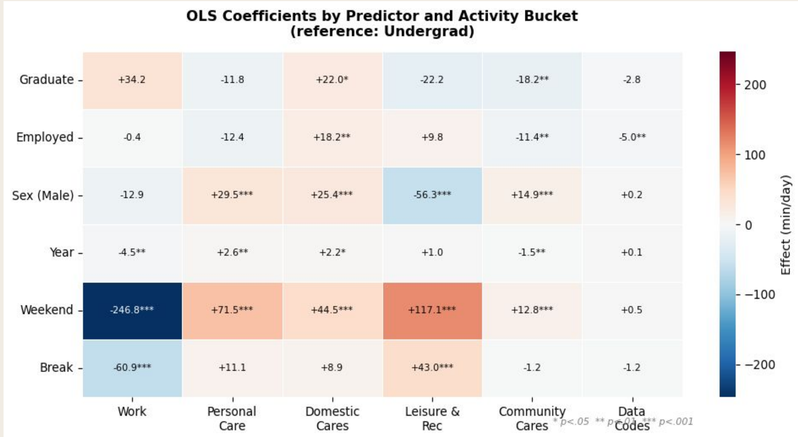


Methodology

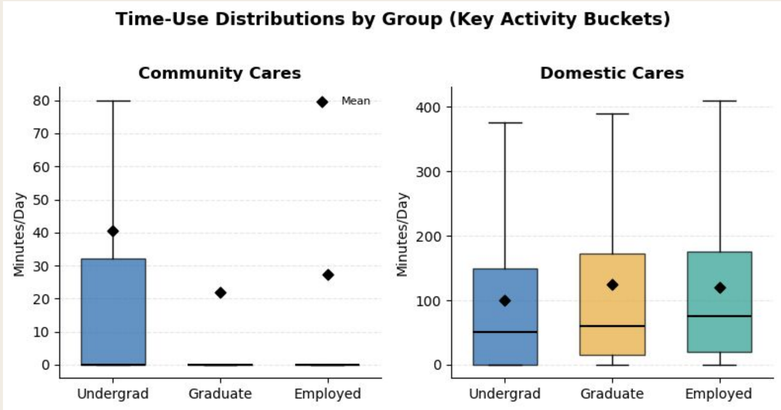
1. Multinomial logistic regressions of group on background vs group on background + activities
2. Likelihood Ratio Test showed adding activities do improve group prediction

Model	Log-Likelihood	AIC
Background only	-1945.46	3910.92
With Activites	-1922.06	3888.12

   - p-value < .01
3. Linear regression of activities ~ group + background



Results



Highlights & Future Work

Goup membership matters most for domestic and community cares.

- Community care: 18min less – grad, 11min less – employ
- Domestic care: 21min more – grad, 18min more – employ

Limitations:

- Time tracking accuracy from recollection is questionable
- Does not consider different job/academia fields
- R^2 was low across all activities (.0046-.2239)

Future Work:

- Investigating the reasons why undergrads commit to more community cares (i.e. volunteering)
- Investigating within field differences (i.e. are medical students the one dragging leisure & rec down?)
- Building an always on automatic time tracking system

Acknowledgements

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References

1. Bureau of Labor Statistics, U.S. Department of Labor. American Time Use Survey, 2024. https://www.bls.gov/tus/